Trial: The Forged Seal

Instructor guide · Req 6c (use a hashing tool to show file integrity) · 12 minutes (follows the Wax Seal slide)

What Req 6c asks (hashing option)

Use a hashing tool such as SHA or MD5 to create a checksum for a file, document or text. Have a fellow Warden or the counselor change it, recreate the checksum and compare the two to show file integrity checking. Every Warden who wants Req 6c credit must do the seal, change and compare steps themselves.

Press the Seal	Spot the Forgery	Debrief
6 min	4 min	2 min
Req 6c demo	case files	

Before class

1) Open **Wax Seal Lab.html** on your laptop and test it on a Mac and a Windows device. It runs offline in any browser (nothing is installed, nothing leaves the device). 2) Get it onto Wardens' devices: email it to families ahead of class, share by AirDrop or USB, or have lodges share one device. 3) Print the worksheet (one per Warden), the case file page (one per lodge), and keep the answer key.

Run it

Before you start: the Wax Seal slide has already taught that a hash is a fingerprint for data that cannot be reversed. **Press the Seal:** each Warden hashes a short message in Wax Seal Lab tab 2, copies the first 8 characters, has a teammate change one character, then compares. Check the worksheet and initial it for Req 6c. **Spot the Forgery:** each lodge turns over its case file, types each document exactly into Wax Seal Lab tab 3, enters the posted checksum and marks Untouched or Tampered. **Debrief:** ask what a mismatch tells you, and why a download site would post a checksum.

Scoring (Cup of the Silver Padlock)

Press the Seal worksheet complete **+2** per lodge · first lodge to find the tampered document **+3** · each other lodge that finds it **+1** · explaining why hashing is not encryption **+1**.

Teach these points

• A hash is **not** encryption. Encryption can be undone with a key; a hash is one-way. • Websites store a hash of your password, not the password. • A matching checksum means the file is the same; it does not prove the file is safe. • MD5 and SHA-1 are old and can be fooled on purpose; SHA-256 is the modern choice. • A single typo gives a mismatch too: check the character count first.

Without Wax Seal Lab (own tools)

For a real file: Windows PowerShell `Get-FileHash file.txt -Algorithm SHA256` · Mac Terminal `shasum -a 256 file.txt`. Either way, compare the two results for the original and the changed copy.

Press the Seal

Warden worksheet · Req 6c · one per Warden

Name: _____ Lodge: ____ Tool used (circle): Wax Seal Lab PowerShell Terminal Other _____

Step 1. Seal it

Write a short, kind message (one sentence, no personal details). Put it in Wax Seal Lab and copy the first 8 characters of the SHA-256 hash.

My message:

_____

_____

Checksum A (first 8 characters): ____ ____ ____ ____ ____ ____ ____ ____

Step 2. A teammate changes one thing

Your lodge mate changes one character (a letter, a space or a punctuation mark) without telling you which one.

The changed message:

_____

_____

Checksum B (first 8 characters): ____ ____ ____ ____ ____ ____ ____ ____

Step 3. Compare

Do A and B match? ☐ Yes ☐ No How many of the 8 characters changed? ____

Find the changed character: _____ What did the checksum tell you?

_____

Hashing vs encryption: one difference is...

_____

_____

Counselor initials for Req 6c: _____ Date: _____

Spot the Forgery

Case files · print once, cut apart, one per lodge · hand out face down

Case File · Lodge 1

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Camp fee is $54 per Warden.` 27 characters	`66addc55`	☐ Untouched ☐ Tampered
2	`Bring your own water bottle.` 28 characters	`0371cd61`	☐ Untouched ☐ Tampered
3	`Leave at 7:30 AM from the church lot.` 37 characters	`3f18bbd6`	☐ Untouched ☐ Tampered
4	`Lights out at 9:30 PM sharp.` 28 characters	`a2c6abd1`	☐ Untouched ☐ Tampered

Case File · Lodge 2

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Meet at the flagpole at noon.` 29 characters	`66a47eef`	☐ Untouched ☐ Tampered
2	`Pack lunch for Saturday only.` 29 characters	`500ee3b8`	☐ Untouched ☐ Tampered
3	`Rain location is the fellowship hail.` 37 characters	`2f66042e`	☐ Untouched ☐ Tampered
4	`Parents may pick up at 4:00 PM.` 31 characters	`87e1d235`	☐ Untouched ☐ Tampered

Case File · Lodge 3

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Wi-Fi password is posted at the lodge.` 38 characters	`55fefe6f`	☐ Untouched ☐ Tampered
2	`Do not share your password with anyone.` 39 characters	`b79dc130`	☐ Untouched ☐ Tampered
3	`Update your phone before the trip.` 34 characters	`3178b8b0`	☐ Untouched ☐ Tampered
4	`Counselor email is on the sign-in sheets.` 41 characters	`f5bfc973`	☐ Untouched ☐ Tampered

Case File · Lodge 4

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Report any scam text to an adult.` 33 characters	`e1dbd040`	☐ Untouched ☐ Tampered
2	`Never click links from stranger.` 32 characters	`f55aea31`	☐ Untouched ☐ Tampered
3	`Turn on two-step login today.` 29 characters	`a14a5bee`	☐ Untouched ☐ Tampered
4	`Back up your photos each month.` 31 characters	`3e0cbe58`	☐ Untouched ☐ Tampered

Case File · Lodge 5

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Troop meets every Tuesday at 7 PM.` 34 characters	`45efdd8c`	☐ Untouched ☐ Tampered
2	`Dues are due on the first meeting.` 34 characters	`af5056be`	☐ Untouched ☐ Tampered
3	`The hike starts at Trail Marker 21.` 35 characters	`f70d9c57`	☐ Untouched ☐ Tampered
4	`Wear closed-toe shoes for the hike.` 35 characters	`fe57673d`	☐ Untouched ☐ Tampered

Case File · Lodge 6

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Safety check: helmets on before riding.` 39 characters	`4c9e6370`	☐ Untouched ☐ Tampered
2	`First aid kit stays with the leader.` 36 characters	`808a8041`	☐ Untouched ☐ Tampered
3	`Our lodge flag is green and gold.` 33 characters	`c76956c3`	☐ Untouched ☐ Tampered
4	`Check the fire ring is cold before leaving!` 43 characters	`eb0a6a57`	☐ Untouched ☐ Tampered

Spot the Forgery: Answer Key and Scoring

Counselor copy · do not hand out

Lodge	Tampered	Text as printed	Posted (original)	Actual (printed text)
Lodge 1	**Doc 1**	Camp fee is $54 per Warden.	**66addc55**	**de282a77**
Lodge 2	**Doc 3**	Rain location is the fellowship hail.	**2f66042e**	**9dfb101f**
Lodge 3	**Doc 4**	Counselor email is on the sign-in sheets.	**f5bfc973**	**3f0ce4aa**
Lodge 4	**Doc 2**	Never click links from stranger.	**f55aea31**	**71df43d5**
Lodge 5	**Doc 3**	The hike starts at Trail Marker 21.	**f70d9c57**	**3856ceb1**
Lodge 6	**Doc 4**	Check the fire ring is cold before leaving!	**eb0a6a57**	**ab746dcf**

How each was changed

P1: $45 → $54 (digits swapped). P2: "hall" → "hail". P3: "sheet" → "sheets". P4: "strangers" → "stranger". P5: Trail Marker 12 → 21. P6: "." → "!". All look nearly identical, which is the point of a checksum.

Note

Each posted checksum is the first 8 characters of the SHA-256 hash of the original text. A Warden who types a document with a typo may see a mismatch even on an untouched document: have them check the character count and retype it.

Lodge	Seal worksheet +2	First fake found +3	Others found +1	Hash vs encryption +1	Total
1					
2					
3					
4					
5					
6					

Sign-off check (Req 6c)

☐ Each Warden created a checksum ☐ A teammate or counselor changed the text ☐ Each Warden recreated the checksum and compared ☐ Tracker updated (6c)